Hedging Personal Income Risk Using Prediction Markets

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Abstract

Income risk is one of the largest sources of financial uncertainty for individuals. This paper proposes a framework for hedging personal job-loss risk using event contracts from prediction markets. Employment is modeled as a stochastic process with a macro-sensitive hazard rate governing job loss events. We construct a hedge portfolio using contracts whose payoffs are positively correlated with unemployment or recession risk. Using Monte Carlo simulations, we evaluate how such hedges reduce the left-tail risk of personal income distributions while preserving expected income.

1 Introduction

Household income risk is typically addressed through savings or traditional insurance products. However, these approaches provide limited protection against macroeconomic shocks that simultaneously increase unemployment risk and reduce future earning opportunities.

Prediction markets such as Kalshi offer tradable contracts whose payouts are tied to macroeconomic events including unemployment levels, recessions, and interest rate changes. These instruments provide a potential mechanism for hedging personal income risk.

This paper models employment risk as a stochastic process and evaluates how prediction market contracts can reduce tail risk in individual income streams.

2 Income Process

Let E_t denote the employment state:

$$E_t = \begin{cases} 1 & \text{if employed} \\ 0 & \text{if unemployed} \end{cases}$$

Let S represent salary per unit time. The income accumulation process is

$$dW_t = SE_t dt$$

where W_t is cumulative wealth from labor income.

3 Job Loss Model

Job loss events are modeled as a stochastic jump process. Let N_t be a Poisson process with intensity λ_t representing the job-loss hazard rate.

$$N_t \sim Poisson(\lambda_t)$$

Employment evolves according to

$$dE_t = -E_t dN_t$$

meaning employment transitions from 1 to 0 when a job-loss event occurs.
The hazard rate depends on macroeconomic variables:

$$\lambda_t = \alpha + \beta_1 U_t + \beta_2 r_t + \beta_3 I_t$$

where

- U_t = unemployment rate
- r_t = interest rate
- I_t = industry-specific risk indicators

4 Income Jump-Diffusion Model

To capture salary variability and sudden job loss shocks, income is modeled as a jump-diffusion process:

$$dW_t = Sdt + \sigma dB_t - JdN_t$$

where

- B_t is Brownian motion representing income variability
- σ is salary volatility
- J is the income loss associated with job loss
- N_t is the job-loss jump process

5 Prediction Market Hedge

Let H_t denote the value of a hedge constructed using prediction market contracts tied to macroeconomic events.

The combined wealth process becomes

$$dW_t = SE_t dt + dH_t$$

The hedge is chosen such that

$$Corr(\text{JobLoss}, H_t) > 0$$

ensuring the hedge pays out when unemployment risk increases.

6 Optimal Hedge Ratio

We determine the optimal hedge ratio using the minimum variance hedge:

$$h^* = \frac{Cov(L, H)}{Var(H)}$$

where

- L is income loss due to unemployment
- H is hedge payoff

The optimal hedge reduces variance while minimizing hedge cost.

7 Model Inputs and Data

The model requires three categories of inputs.

7.1 Personal Risk Factors

These estimate the individual's job-loss probability. Inputs include: industry, company size, job level (entry, manager, exec), tenure, salary, Monte Carlo path count, and horizon years. These affect the baseline hazard rate $\lambda(t)$.

7.2 Macroeconomic Variables

These drive layoffs and hiring cycles. The state vector is

$$X_t = (U_t, r_t, \text{GDP}_t, \text{Layoffs}_t)$$

where U_t is unemployment rate, r_t is the interest rate, GDP_t is growth, and Layoffs_t is industry layoff activity. Data sources:

- **FRED** (Federal Reserve): unemployment (UNRATE), initial jobless claims (ICSA), federal funds rate (FEDFUNDS), GDP growth
- **BLS JOLTS**: layoffs (JTSLDL), hires (JTSHIR)

7.3 Market Expectations (Forward-Looking Risk)

Sources: Kalshi API, Polymarket, Fed futures (CME FedWatch), yield curve. Markets contain implied probabilities: $P(\text{recession})$, $P(\text{unemployment} > x)$. FRED provides the 10Y-2Y Treasury spread (T10Y2Y) as a recession indicator; inversion historically precedes recessions.

7.4 Hedge Market Variables

From Kalshi or prediction markets: probability unemployment $> 5.5\%$, recession probability, interest rate changes, CPI surprises. These determine hedge payoff structure.

7.5 Salary Reference Data

For calibration: BLS wage tables, Glassdoor, Levels.fyi (tech). Salary distribution by job level informs income loss magnitude.

7.6 Unemployment Duration

From BLS (via FRED): average weeks unemployed (UEMPMEAN). Typical mean duration ≈ 20 weeks. Used for reemployment hazard modeling.

8 Hazard Model for Job Loss

We estimate job-loss probability via logistic regression:

$$P(\text{JobLoss}) = \sigma\left(\beta_0 + \beta_1 U_t + \beta_2 r_t + \beta_3 I_t\right)$$

where

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

and U_t is unemployment, r_t is the interest rate, and I_t is industry risk. Training uses historical layoffs and macro variables. The weekly Poisson intensity λ is derived from the monthly probability.

Example coefficient output used in documentation:

$$\alpha = 0.002, \quad \beta_1 = 0.04, \quad \beta_2 = 0.01, \quad \beta_3 = 0.03$$

9 Stochastic Model

Once λ is estimated: $N_t \sim \text{Poisson}(\lambda t)$. Job loss occurs when a Poisson jump happens. Employment: $E_t = 1$ if employed, $E_t = 0$ if unemployed. Income: $dW_t = SE_t\, dt$.

10 Hedge Payoff Model

Kalshi contracts are binary: payoff \$1 if event occurs, \$0 otherwise. Example: "Unemployment $> 6\%$" with market price 0.30 implies 30% probability. Hedge value: $H = \text{contract_price} \times \text{contracts}$.

11 Monte Carlo Simulation

We simulate thousands of career paths (default: 5,000 paths, 10-year horizon, weekly time steps). For each path:

1. Simulate macro environment (unemployment path)
2. Compute job-loss hazard $\lambda(t)$ from macro state
3. Simulate Poisson event (job loss)
4. Compute salary path $dW_t = SE_t\, dt$
5. Apply hedge payoff

11.1 Default Dashboard Configuration

For the default dashboard profile (Tech, Startup, Entry):

$$S = 120000, \quad N = 5000, \quad T = 10$$

and Monte Carlo generates

$$W_t^{(1)}, W_t^{(2)}, \ldots, W_t^{(5000)}.$$

12 Risk Metrics

We evaluate:

- **Variance**: income volatility
- **Expected Shortfall**: average loss in worst 5% of cases
- **Tail probability**: $P(\text{income drop} > 50\%)$

Goal: reduce tail risk while preserving expected income.

13 Optimal Hedge Ratio

Choose hedge size using the minimum-variance hedge:

$$h^* = \frac{\text{Cov}(\text{Loss}, \text{Hedge})}{\text{Var}(\text{Hedge})}$$

This minimizes the variance of (Loss $-h\cdot$ Hedge).

14 System Pipeline

personal data
↓
hazard model
↓
job-loss simulation (Poisson)
↓
Monte Carlo simulation
↓
income distribution
optimal hedge ratio
↓
hedge portfolio

15 Discussion

Prediction markets provide a novel mechanism for hedging personal income risk. Unlike traditional insurance, these markets allow individuals to dynamically hedge macroeconomic shocks that increase unemployment risk.

The proposed framework combines hazard modeling, stochastic income processes, and derivative-style hedging to reduce tail risk in individual wealth distributions.

Future work could incorporate industry-specific job-loss models, copula-based correlation structures, and real-time hedge optimization using market-implied probabilities.

16 Conclusion

This paper presents a stochastic framework for hedging job-loss risk using prediction market contracts. By modeling employment as a macro-sensitive hazard process and constructing correlated hedges, individuals can reduce the downside risk of income shocks while maintaining expected earnings.